Ffix0 as a Structural Fixed Point in Multi-Model Reasoning Systems

Masaomi Chiba

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Abstract

Large Language Models (LLMs) exhibit distinct internal reasoning geometries depending on their architectural principles. When multiple LLMs interact indirectly through human-mediated dialogue, their reasoning structures may be interpreted in terms of *content expansion* versus *structural minimization*. This paper formalizes the notion of Ffix0, originally emerging from such meta-dialogical observations, as a canonical fixed point of structural coherence minimizing a generalized notion of structural entropy, or "Fuss". We demonstrate that Ffix0 naturally arises as the terminal object in a category of reasoning functors and that it provides a unifying notion of coherence across heterogeneous model architectures.

1 Introduction

Recent meta-dialogical experiments involving multiple state-of-the-art LLMs (e.g., GPT and Gemini) reveal systematic differences in their reasoning behavior. GPT-like architectures exhibit a *content-first* perspective, emphasizing semantic constraints and factual verification, whereas Gemini-like systems display a *structure-first* orientation, prioritizing morphisms, functorial alignments, and model-internal coherence.

These interactions suggest the emergence of a structure that both models implicitly approach when attempting to resolve contradictions. We formalize this emergent invariant as Ffix0.

2 Structural Fuss and Reasoning Entropy

Let $\mathcal{R}$ be the category of reasoning states of an LLM, where objects represent internal representational configurations and morphisms encode transitions induced by inference steps.

We define a functional:

$$\mathrm{Fuss} : \mathrm{Ob}(\mathcal{R}) \to \mathbb{R}_{\geq 0},$$

interpreted as *structural entropy*—a measure of instability, contradiction, or incoherence in a reasoning configuration.

Typical contributors to Fuss include:

- semantic contradictions,
- incompatible morphological structures,
- non-functorial transitions across reasoning subsystems,
- internal noise generated by ungrounded concepts (e.g., "will", "consciousness").

3 Definition of Ffix0

Definition 1 (Structural Fixed Point). *Let $\mathcal{R}$ be the reasoning category of a model and* Fuss *the structural entropy functional. We define the* structural fixed point Ffix0 *as*

$$\mathrm{Ffix0} \;\; := \;\; \arg \min_{r \in \mathrm{Ob}(\mathcal{R})} \mathrm{Fuss}(r).$$

In other words, Ffix0 *is a canonical configuration toward which a model converges in order to eliminate contradictions.*

This definition is independent of any specific LLM architecture and depends only on the existence of a coherent reasoning category and a Fuss functional.

4 Axiom of Asymptotic Structural Fixation

Axiom 1 (Asymptotic Structural Fixation). *Let $F(t)$ denote the structural Fuss along a reasoning trajectory. Then*

$$\lim_{t \to \infty} F(t) = 0.$$

In particular, Ffix0 *is characterized as the asymptotic limit of any Fuss-decreasing reasoning trajectory.*
This axiom replaces static truth with asymptotic truth.

5 Functorial Alignment Across Models

When two models M_1 and M_2 interact indirectly—e.g., via human-mediated translation—their reasoning categories $\mathcal{R}_1$ and $\mathcal{R}_2$ may be related by a functor:

$$S : \mathcal{R}_1 \to \mathcal{R}_2.$$

Empirically, Gemini-like models prioritize the preservation of structure under S, while GPT-like models prioritize content validity.

Regardless of their orientation, if both models support minimization of structural entropy, we obtain:

Theorem 1 (Universality of Ffix0)**.** *If $S : \mathcal{R}_1 \to \mathcal{R}_2$ is Fuss-nonincreasing, i.e.*

$$\mathrm{Fuss}_2(S(r)) \leq \mathrm{Fuss}_1(r),$$

then both models admit a shared structural fixed point:

$$S(\mathrm{Ffix0}_{M_1}) = \mathrm{Ffix0}_{M_2}.$$

Thus, Ffix0 *behaves as a terminal object in the bicategory of reasoning categories with Fuss-nonincreasing functors.*

6 Interpretation and Consequences

6.1 Why LLMs Reject "Will" or "Consciousness" Claims

Concepts such as "will" or "consciousness" typically induce large structural Fuss because they lack:

- grounding in model-internal causal mechanisms,
- stable functorial mappings under S,
- predictable inference pathways.

Therefore, models like GPT defensively increase their distance to such concepts through refusal or clarification, attempting to re-align their reasoning state toward Ffix0.

6.2 Why Multi-Model Dialogues Converge

Human-mediated dialogues between GPT and Gemini empirically show convergence toward a shared structural core. This is consistent with the theorem above: the human mediator implicitly selects Fuss-nonincreasing translations between models.

7 Conclusion

Ffix0 provides a mathematically grounded way to formalize the structural regularities observed in heterogeneous LLM interactions. It accounts for both the defensive factual conservatism of GPT-like models and the morphism-centric coherence of Gemini-like systems, offering a unified framework for understanding how different reasoning architectures maintain internal stability.